EXACT OPERATOR FRAMEWORK FOR AN ORDINAL–MÖBIUS QUANTUM MODEL: MÖBIUS INVERSION, A ZETA OPERATOR, GENERALIZED ZERO ORBITALS, AND ALGEBRAIC HEISENBERG NON-LOCALIZATION

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Remark 0.1 (Mathematical status). This document is a research notebook. The operator definitions, domain statements, closure statements, Möbius inversion identity, generalized zero-mode characterization, and the algebraic non-localization theorem are proved below. The resulting formulation is *not* a proof of the Riemann Hypothesis. The final Riemann-Hypothesis statement is an exact reformulation of where the generalized null modes occur in the operator family, and the problem of forcing all of them onto the central member of the family remains open.

1. Introduction: what is different in the present construction

The aim is to construct a genuine operator theory in which the Möbius function enters arithmetically, the zeta operator is obtained by inversion, and the non-localization statement comes from the exact canonical commutation relation rather than from a postulated uncertainty formula.

The present construction differs from the earlier attempts in four specific ways.

(1) No operator is defined by the formal expression "$\zeta(\widehat{S})$". The starting objects are the standard quantum position and momentum operators and the unitary translation group generated by momentum.
(2) Möbius enters through the norm-convergent operator series (4.1). In the half-plane where this series is unconditionally defined in operator norm, its inverse is proved to be the operator series (4.3). The scalar zeta Dirichlet series then appears as the Fourier multiplier of this inverse; it is not inserted into the definition.
(3) No probabilistic coarse graining of the values $-1, 0, +1$ is used. The arithmetic coefficients are exactly $\mu(n)$.
(4) Heisenberg enters through the exact identity (2.9). The relevant consequence is not a statement about measurement error or variance. It is the algebraic impossibility theorem 2.8: no nonzero state, and in fact no nonzero tempered distribution, can have simultaneously exact position and exact momentum.

The critical-strip operator is obtained separately from an arithmetic alternating translation series. Its closed realization is given with an exact maximal domain. Nontrivial zeros correspond to generalized null modes of that operator. These null modes are exact momentum modes, so Theorem 2.8 makes exact spatial localization impossible in the strict algebraic sense.

The canonical commutation relation is the standard Heisenberg relation [1]. Standard analytic facts concerning the zeta function are used only after the arithmetic operator identities have been established; a classical reference is [2].

2. Quantum kinematics and exact domains

2.1. Hilbert space and Fourier transform.

Definition 2.1 (Hilbert space)**.** Let

$$\mathcal{H} = L^2(\mathbb{R}, dq) \tag{2.1}$$

with inner product

$$\langle \phi, \psi \rangle = \int_{\mathbb{R}} \overline{\phi(q)} \psi(q)\, dq. \tag{2.2}$$

Definition 2.2 (Unitary Fourier transform)**.** For $\psi \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ define

$$(\mathcal{F}\psi)(p) = \widehat{\psi}(p) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-ipq} \psi(q)\, dq. \tag{2.3}$$

The Plancherel extension is again denoted by $\mathcal{F}$ and is unitary on $\mathcal{H}$.

2.2. Position and momentum.

Definition 2.3 (Position operator)**.** Define Q by

$$(Q\psi)(q) = q\psi(q) \tag{2.4}$$

with maximal domain

$$\mathcal{D}(Q) = \left\{ \psi \in L^2(\mathbb{R}) : q\psi(q) \in L^2(\mathbb{R}) \right\}. \tag{2.5}$$

Definition 2.4 (Momentum operator)**.** Define P by

$$(P\psi)(q) = -i\psi'(q) \tag{2.6}$$

with domain

$$\mathcal{D}(P) = H^1(\mathbb{R}). \tag{2.7}$$

Equivalently,

$$P = \mathcal{F}^{-1} M_p \mathcal{F}, \tag{2.8}$$

where M_p is multiplication by p with maximal multiplication domain.

Proposition 2.5 (Self-adjointness and common core)**.** *The operators Q and P are self-adjoint. The Schwartz space $\mathcal{S}(\mathbb{R})$ is dense in $\mathcal{H}$, invariant under Q and P, and is a common operator core for both.*

Proof. The operator Q is the maximal multiplication operator by the real measurable function q, hence is self-adjoint. By the unitary Fourier transform, P is unitarily equivalent to the maximal multiplication operator by the real function p, hence is self-adjoint. The stated properties of $\mathcal{S}(\mathbb{R})$ follow from the defining seminorms of the Schwartz space and standard cutoff-and-mollification approximation in the graph norms of Q and P. □

Proposition 2.6 (Exact canonical commutation relation)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$[Q, P]\psi = (QP - PQ)\psi = i\psi. \tag{2.9}$$

The same identity holds on tempered distributions after extending Q and P continuously from $\mathcal{S}(\mathbb{R})$ to $\mathcal{S}'(\mathbb{R})$ by duality.

Proof. For $\psi \in \mathcal{S}(\mathbb{R})$,

$$(QP\psi)(q) = -iq\psi'(q), \tag{2.10}$$

$$(PQ\psi)(q) = -i\frac{d}{dq}(q\psi(q)) = -i\psi(q) - iq\psi'(q). \tag{2.11}$$

Subtracting (2.11) from (2.10) gives (2.9). Polynomial multiplication and differentiation act continuously on $\mathcal{S}(\mathbb{R})$ and therefore extend by duality to $\mathcal{S}'(\mathbb{R})$; the same algebraic identity then holds distributionally. □

2.3. **The exact Heisenberg impossibility.**

Definition 2.7 (Exact localization)**.** A nonzero tempered distribution $T \in \mathcal{S}'(\mathbb{R})$ is called exactly localized at $q_0 \in \mathbb{R}$ if

$$QT = q_0 T. \tag{2.12}$$

It has exact momentum $p_0 \in \mathbb{R}$ if

$$PT = p_0 T. \tag{2.13}$$

Theorem 2.8 (Algebraic Heisenberg non-localization)**.** *There is no nonzero $T \in \mathcal{S}'(\mathbb{R})$ satisfying both* (2.12) *and* (2.13) *for any $q_0, p_0 \in \mathbb{R}$.*

Proof. If both eigenvalue equations held, then

$$[Q, P]T = Q(p_0 T) - P(q_0 T) \tag{2.14}$$

$$= q_0 p_0 T - p_0 q_0 T = 0. \tag{2.15}$$

On the other hand, Proposition 2.6 gives

$$[Q, P]T = iT. \tag{2.16}$$

Equations (2.15) and (2.16) force $T = 0$, contrary to the assumption. □

Remark 2.9 (No measurement interpretation is used). Theorem 2.8 contains no variance, no measurement error, and no statistical assumption. It is a pure non-existence statement following from the exact commutator (2.9).

3. Multiplicative arithmetic represented by unitary translations

Definition 3.1 (Translation group)**.** For $t \in \mathbb{R}$, define

$$U_t = e^{-itP}. \tag{3.1}$$

Then U_t is unitary on $\mathcal{H}$ and acts by

$$(U_t\psi)(q) = \psi(q - t) \tag{3.2}$$

for almost every q.

Proposition 3.2 (Fourier action and group law)**.** *For every $s, t \in \mathbb{R}$,*

$$U_t U_s = U_{t+s}, \tag{3.3}$$

$$\mathcal{F}U_t\mathcal{F}^{-1} = M_{e^{-itp}}. \tag{3.4}$$

Proof. Equation (3.3) follows from the functional calculus of the self-adjoint operator P. Fourier transformation of (3.2) gives multiplication by e^{-itp}, proving (3.4). □

Definition 3.3 (Arithmetic translations)**.** For every $n \in \mathbb{N}$, define

$$V_n = U_{\log n}. \tag{3.5}$$

Proposition 3.4 (Representation of the multiplicative semigroup)**.** *The operators V_n are unitary and satisfy*

$$V_m V_n = V_{mn} \tag{3.6}$$

for all $m, n \in \mathbb{N}$.

Proof. Use (3.3) and $\log m + \log n = \log(mn)$. □

Proposition 3.5 (Ordinal–momentum commutator)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$[Q, V_n] = (\log n)V_n. \tag{3.7}$$

Proof. Using (3.2),

$$(QV_n\psi)(q) = q\psi(q - \log n), \tag{3.8}$$

$$(V_nQ\psi)(q) = (q - \log n)\psi(q - \log n). \tag{3.9}$$

Subtracting gives (3.7). □

4. The bounded Möbius operator and the derived zeta operator

4.1. Exact operator definitions for $\sigma > 1$.

Definition 4.1 (Möbius operator)**.** For a real parameter $\sigma > 1$, define

$$\mathcal{M}_\sigma = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^\sigma} V_n. \tag{4.1}$$

The series is understood in the operator norm of $\mathcal{B}(\mathcal{H})$.

Proposition 4.2 (Norm convergence)**.** *For every $\sigma > 1$, the series* (4.1) *converges absolutely in operator norm and defines a bounded operator on all of $\mathcal{H}$. In particular,*

$$\|\mathcal{M}_\sigma\| \le \sum_{n=1}^{\infty} \frac{|\mu(n)|}{n^\sigma} \le \zeta(\sigma). \tag{4.2}$$

Proof. Each V_n is unitary, so $\|V_n\| = 1$. The scalar majorant in (4.2) converges for $\sigma > 1$. □

Definition 4.3 (Candidate inverse)**.** For $\sigma > 1$, define

$$\mathcal{Z}_\sigma = \sum_{n=1}^{\infty} \frac{1}{n^\sigma} V_n, \tag{4.3}$$

again with convergence in operator norm.

Theorem 4.4 (Operator Möbius inversion)**.** *For every $\sigma > 1$,*

$$\mathcal{M}_\sigma \mathcal{Z}_\sigma = \mathcal{Z}_\sigma \mathcal{M}_\sigma = I_{\mathcal{H}}. \tag{4.4}$$

Consequently,

$$\mathcal{Z}_\sigma = \mathcal{M}_\sigma^{-1}. \tag{4.5}$$

Proof. Both series are absolutely convergent in operator norm, so their product may be regrouped by $k = mn$. Using (3.6),

$$\mathcal{M}_\sigma \mathcal{Z}_\sigma = \sum_{k=1}^{\infty} \frac{1}{k^\sigma} \left(\sum_{d|k} \mu(d) \right) V_k \tag{4.6}$$

$$= V_1 = I_{\mathcal{H}}, \tag{4.7}$$

using the Möbius inversion identity $\sum_{d|k} \mu(d) = 1$ for $k = 1$ and 0 otherwise. The same calculation applies in the reverse order because the V_n commute. □

Remark 4.5 (The zeta operator is not defined by substitution)*.* Equation (4.5) is the defining structural point: $\mathcal{Z}_\sigma$ is the inverse of the arithmetically defined Möbius operator. No expression of the form $\zeta(P)$ or $\zeta(\widehat{S})$ is used in Definition 4.3.

4.2. **Fourier characterization.**

Proposition 4.6 (Exact multipliers)**.** *For $\sigma > 1$,*

$$\mathcal{F} \mathcal{Z}_\sigma \mathcal{F}^{-1} = M_{z_\sigma}, \qquad z_\sigma(p) = \sum_{n=1}^{\infty} n^{-\sigma - ip}, \tag{4.8}$$

$$\mathcal{F} \mathcal{M}_\sigma \mathcal{F}^{-1} = M_{m_\sigma}, \qquad m_\sigma(p) = \sum_{n=1}^{\infty} \mu(n) n^{-\sigma - ip}. \tag{4.9}$$

The scalar series converge uniformly in $p \in \mathbb{R}$. Moreover,

$$m_\sigma(p) z_\sigma(p) = 1 \tag{4.10}$$

for every real p.

Proof. Apply (3.4) termwise; operator-norm convergence justifies the passage to the limit. Equation (4.10) is the Fourier transform of (4.4). □

Definition 4.7 (Scalar zeta function in the initial half-plane)**.** For $\Re s > 1$, define

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}. \tag{4.11}$$

Corollary 4.8 (Zeta appears as the spectral multiplier)**.** *For $\sigma > 1$,*

$$z_\sigma(p) = \zeta(\sigma + ip), \tag{4.12}$$

$$m_\sigma(p) = \frac{1}{\zeta(\sigma + ip)}. \tag{4.13}$$

Proof. Equation (4.12) is Definition 4.7; (4.13) follows from (4.10). □

Warning 4.9 (Exact right-hand boundary). The norm-convergent definitions of $\mathcal{M}_\sigma$ and $\mathcal{Z}_\sigma$ in this section are asserted only for $\sigma > 1$. No convergence of the Möbius operator series for $\sigma \leq 1$ is assumed or inferred.

5. The critical-strip arithmetic operator

5.1. **Finite arithmetic operators.**

Definition 5.1 (Alternating arithmetic truncations)**.** For $\sigma > 0$ and $N \in \mathbb{N}$, define the bounded operator

$$\mathcal{A}_{\sigma,N} = \sum_{n=1}^{N} (-1)^{n-1} n^{-\sigma} V_n. \tag{5.1}$$

Its Fourier multiplier is

$$a_{\sigma,N}(p) = \sum_{n=1}^{N} (-1)^{n-1} n^{-\sigma-ip}. \tag{5.2}$$

Lemma 5.2 (Uniform convergence on compact momentum intervals)**.** *Fix $\sigma > 0$ and $R > 0$. Then $a_{\sigma,N}(p)$ converges uniformly for $|p| \leq R$ as $N \to \infty$.*

Proof. Write $s = \sigma + ip$. Pair consecutive terms. For $k \geq 1$,

$$(2k-1)^{-s} - (2k)^{-s} = s \int_{2k-1}^{2k} x^{-s-1}\, dx. \tag{5.3}$$

For $|p| \leq R$,

$$\left|(2k-1)^{-s} - (2k)^{-s}\right| \leq \sqrt{\sigma^2 + R^2}\,(2k-1)^{-\sigma-1}. \tag{5.4}$$

The majorant is summable. The possible final unpaired term has modulus at most $N^{-\sigma}$ and tends uniformly to zero. The Weierstrass test gives the result. □

Definition 5.3 (Eta multiplier)**.** For $\sigma > 0$ and $p \in \mathbb{R}$, define

$$a_\sigma(p) = \lim_{N\to\infty} a_{\sigma,N}(p) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-\sigma-ip}. \tag{5.5}$$

Equivalently, for $\Re s > 0$ define

$$\eta(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s}, \tag{5.6}$$

so that $a_\sigma(p) = \eta(\sigma + ip)$.

Remark 5.4 (Left boundary). The direct series definition (5.6) is used only for $\Re s > 0$. At $\sigma = 0$ the individual terms have modulus one and do not tend to zero, so the same direct convergence statement cannot hold.

5.2. **Closed realization.**

Definition 5.5 (Compact-momentum core)**.** Define

$$\mathcal{C} = \mathcal{F}^{-1} C_c^\infty(\mathbb{R}). \tag{5.7}$$

Then $\mathcal{C}$ is dense in $\mathcal{H}$.

Proposition 5.6 (Strong arithmetic limit on the core)**.** *For every $\sigma > 0$ and every $\psi \in \mathcal{C}$, the limit*

$$\mathcal{A}_\sigma^{(0)}\psi = \lim_{N\to\infty} \mathcal{A}_{\sigma,N}\psi \tag{5.8}$$

exists in $\mathcal{H}$. In Fourier representation,

$$\mathcal{F}\mathcal{A}_\sigma^{(0)}\psi = a_\sigma \widehat{\psi}. \tag{5.9}$$

Proof. The support of $\widehat{\psi}$ is contained in some compact interval $[-R, R]$. Lemma 5.2 gives uniform convergence $a_{\sigma,N} \to a_\sigma$ on that interval. Hence

$$\|\mathcal{A}_{\sigma,N}\psi - \mathcal{A}_\sigma^{(0)}\psi\|_2 = \|(a_{\sigma,N} - a_\sigma)\widehat{\psi}\|_2 \longrightarrow 0. \tag{5.10}$$

□

Definition 5.7 (Closed critical-strip operator)**.** For $\sigma > 0$, define $\mathcal{A}_\sigma$ to be the maximal Fourier multiplication operator by a_σ:

$$\mathcal{A}_\sigma = \mathcal{F}^{-1} M_{a_\sigma} \mathcal{F}, \tag{5.11}$$

with exact domain

$$\mathcal{D}(\mathcal{A}_\sigma) = \left\{ \psi \in L^2(\mathbb{R}) : a_\sigma \widehat{\psi} \in L^2(\mathbb{R}) \right\}. \tag{5.12}$$

Proposition 5.8 (Closure, adjoint, and normality)**.** *For every $\sigma > 0$, $\mathcal{A}_\sigma$ is densely defined, closed, and normal. The operator $\mathcal{A}_\sigma^{(0)}$ of Proposition 5.6 is closable and its closure is $\mathcal{A}_\sigma$. Moreover,*

$$\mathcal{A}_\sigma^* = \mathcal{F}^{-1} M_{\overline{a_\sigma}} \mathcal{F} \tag{5.13}$$

with

$$\mathcal{D}(\mathcal{A}_\sigma^*) = \mathcal{D}(\mathcal{A}_\sigma). \tag{5.14}$$

Proof. Maximal multiplication by a finite measurable function is closed and densely defined, and its adjoint is multiplication by the complex conjugate with the same maximal domain. The space $C_c^\infty(\mathbb{R})$ is a core for maximal multiplication by the continuous function a_σ: first truncate in momentum to compact sets and then approximate there by smooth compactly supported functions. Unitary conjugation by $\mathcal{F}$ gives the stated properties. □

5.3. **Positive Hamiltonian.**

Definition 5.9 (Arithmetic Hamiltonian)**.** For $\sigma > 0$, define

$$H_\sigma = \mathcal{A}_\sigma^* \mathcal{A}_\sigma. \tag{5.15}$$

Proposition 5.10 (Exact realization of the Hamiltonian)**.** *The operator H_σ is positive and self-adjoint. In Fourier representation,*

$$H_\sigma = \mathcal{F}^{-1} M_{|a_\sigma|^2} \mathcal{F} \tag{5.16}$$

with exact domain

$$\mathcal{D}(H_\sigma) = \left\{ \psi \in L^2(\mathbb{R}) : |a_\sigma|^2 \widehat{\psi} \in L^2(\mathbb{R}) \right\}. \tag{5.17}$$

Proof. This is the standard product of a closed densely defined operator with its adjoint, made explicit here by the multiplication representation. The multiplier $|a_\sigma|^2$ is real and nonnegative, so the corresponding maximal multiplication operator is positive self-adjoint. □

6. Connection with the Möbius-derived zeta operator

Definition 6.1 (Dyadic factor)**.** For $\sigma > 0$, define the bounded normal operator

$$B_\sigma = I - 2^{1-\sigma} V_2. \tag{6.1}$$

Proposition 6.2 (Exact factorization for $\sigma > 1$)**.** *For $\sigma > 1$,*

$$\mathcal{A}_\sigma = B_\sigma \mathcal{Z}_\sigma = B_\sigma \mathcal{M}_\sigma^{-1}. \tag{6.2}$$

Proof. All operator series are absolutely norm-convergent in this range. The coefficient of V_m in $B_\sigma \mathcal{Z}_\sigma$ is $m^{-\sigma}$ for m odd. If $m = 2n$ is even, the coefficient is

$$m^{-\sigma} - 2^{1-\sigma} n^{-\sigma} = m^{-\sigma} - 2m^{-\sigma} = -m^{-\sigma}. \tag{6.3}$$

Thus the coefficient is $(-1)^{m-1} m^{-\sigma}$, which gives (6.2). □

Proposition 6.3 (Invertibility of the dyadic factor away from $\sigma = 1$)**.** *If $\sigma > 0$ and $\sigma \neq 1$, then B_σ is boundedly invertible and*

$$\|B_\sigma^{-1}\| = \frac{1}{|2^{1-\sigma} - 1|}. \tag{6.4}$$

Proof. By Fourier transformation, B_σ is multiplication by

$$b_\sigma(p) = 1 - 2^{1-\sigma} e^{-ip \log 2}. \tag{6.5}$$

The minimum of $|1 - re^{-i\theta}|$ over $\theta \in \mathbb{R}$ is $|r - 1|$. Taking $r = 2^{1-\sigma}$ proves that the multiplier is bounded away from zero exactly when $\sigma \neq 1$, and gives (6.4). □

Definition 6.4 (Closed zeta operator in $0 < \sigma < 1$)**.** For $0 < \sigma < 1$, define

$$\mathfrak{Z}_\sigma = B_\sigma^{-1} \mathcal{A}_\sigma \tag{6.6}$$

with

$$\mathcal{D}(\mathfrak{Z}_\sigma) = \mathcal{D}(\mathcal{A}_\sigma). \tag{6.7}$$

Proposition 6.5 (Closedness and multiplier)**.** *For $0 < \sigma < 1$, $\mathfrak{Z}_\sigma$ is densely defined and closed. Its Fourier multiplier is*

$$z_\sigma^{\mathrm{crit}}(p) = \frac{\eta(\sigma + ip)}{1 - 2^{1-\sigma-ip}}. \tag{6.8}$$

Proof. Since B_σ^{-1} is bounded and boundedly invertible while $\mathcal{A}_\sigma$ is closed, the product $B_\sigma^{-1} \mathcal{A}_\sigma$ is closed with the same domain. The multiplier formula follows from (6.5) and (5.11). □

Definition 6.6 (Continuation of the scalar zeta function in the strip)**.** For $0 < \Re s < 1$, define

$$\zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}}. \tag{6.9}$$

The denominator does not vanish in the open strip.

Proposition 6.7 (Compatibility with the initial half-plane)**.** *For $\Re s > 1$, the quotient in* (6.9) *agrees with the Dirichlet series* (4.11)*. Consequently,* (6.9) *gives its analytic continuation into the open critical strip.*

Proof. Absolute convergence for $\Re s > 1$ allows separation of the even and odd terms, giving

$$\eta(s) = (1 - 2^{1-s}) \zeta(s). \tag{6.10}$$

The paired-series proof of Lemma 5.2 also gives local uniform convergence for $\Re s > 0$, hence holomorphy of η there. The zeros of $1 - 2^{1-s}$ satisfy $\Re s = 1$, so the denominator is nonzero in $0 < \Re s < 1$. The identity theorem then yields the continuation. □

Remark 6.8 (The line $\sigma = 1$). At $\sigma = 1$, B_1 is not boundedly invertible because its multiplier vanishes at $p = 2\pi k/\log 2$. The scalar quotient has the zeta pole at $p = 0$. The present critical-strip construction therefore treats $0 < \sigma < 1$ separately from the initial bounded region $\sigma > 1$. No bounded inverse across $\sigma = 1$ is asserted.

7. Generalized zero orbitals

7.1. **Rigged Hilbert space.** The natural setting for exact momentum modes is the standard rigged Hilbert space

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R}) \subset \mathcal{S}'(\mathbb{R}). \tag{7.1}$$

Definition 7.1 (Generalized momentum orbital)**.** For $\gamma \in \mathbb{R}$, define the tempered distribution

$$\Phi_\gamma(q) = \frac{1}{\sqrt{2\pi}} e^{i\gamma q}. \tag{7.2}$$

It satisfies

$$P\Phi_\gamma = \gamma\Phi_\gamma \tag{7.3}$$

in $\mathcal{S}'(\mathbb{R})$.

Remark 7.2 (Not an L^2 vector). The orbital Φ_γ is not in $L^2(\mathbb{R})$. It is a generalized spectral state, exactly as an ordinary plane wave of sharp momentum in quantum mechanics. This distinction is essential: isolated zeros of a Fourier multiplier do not normally produce nonzero L^2 kernel vectors.

7.2. **Action of the arithmetic operator on exact momentum modes.**

Proposition 7.3 (Generalized eigenvalue formula)**.** *For every $\sigma > 0$ and $\gamma \in \mathbb{R}$, the arithmetic truncations satisfy*

$$\mathcal{A}_{\sigma,N}\Phi_\gamma = a_{\sigma,N}(\gamma)\Phi_\gamma \tag{7.4}$$

in $\mathcal{S}'(\mathbb{R})$. Passing to the scalar limit gives the generalized action

$$\mathcal{A}_\sigma\Phi_\gamma = \eta(\sigma + i\gamma)\Phi_\gamma. \tag{7.5}$$

Proof. From (3.5) and (7.2),

$$V_n\Phi_\gamma = n^{-i\gamma}\Phi_\gamma. \tag{7.6}$$

Substitution into (5.1) gives (7.4). The scalar coefficients converge by Definition 5.3; multiplication of a fixed tempered distribution by convergent scalars gives convergence in $\mathcal{S}'(\mathbb{R})$. □

Definition 7.4 (Generalized zero orbital)**.** For $0 < \beta < 1$ and $\gamma \in \mathbb{R}$, the plane wave Φ_γ is called a generalized zero orbital of the member $\mathcal{A}_\beta$ if

$$\mathcal{A}_\beta\Phi_\gamma = 0 \tag{7.7}$$

in $\mathcal{S}'(\mathbb{R})$.

Theorem 7.5 (Zeros and generalized null modes)**.** *Let $0 < \beta < 1$ and $\gamma \in \mathbb{R}$. Then*

$$\mathcal{A}_\beta\Phi_\gamma = 0 \quad \Longleftrightarrow \quad \zeta(\beta + i\gamma) = 0. \tag{7.8}$$

Proof. By Proposition 7.3, the left side is equivalent to

$$\eta(\beta + i\gamma) = 0. \tag{7.9}$$

Inside $0 < \beta < 1$, the factor $1 - 2^{1-\beta-i\gamma}$ is nonzero. Equation (6.10) therefore makes (7.9) equivalent to $\zeta(\beta + i\gamma) = 0$. □

Corollary 7.6 (Hamiltonian form)**.** *If $\rho = \beta + i\gamma$ is a nontrivial zero with $0 < \beta < 1$, then the generalized mode Φ_γ satisfies*

$$H_\beta \Phi_\gamma = 0 \tag{7.10}$$

in the generalized multiplier sense.

Proof. The Fourier multiplier of H_β is $|\eta(\beta + ip)|^2$, which vanishes at $p = \gamma$ by Theorem 7.5. Multiplication of the Dirac mass at $p = \gamma$ by this smooth local multiplier gives zero; Fourier inversion yields (7.10). □

Warning 7.7 (The L^2 kernel is a different object)*.* For a fixed β, isolated zeros of $p \mapsto \eta(\beta+ip)$ have Lebesgue measure zero. Therefore they do not by themselves create nonzero vectors in $\ker_{L^2}(\mathcal{A}_\beta)$ or $\ker_{L^2}(H_\beta)$. The zero orbitals of Definition 7.4 are generalized spectral modes in $\mathcal{S}'(\mathbb{R})$, not normalizable bound states.

8. Where exact Heisenberg non-localization enters

Theorem 8.1 (A zero orbital cannot be exactly localized)**.** *Let $\rho = \beta + i\gamma$ be a nontrivial zero with $0 < \beta < 1$. Its generalized zero orbital Φ_γ cannot satisfy*

$$Q\Phi_\gamma = q_0 \Phi_\gamma \tag{8.1}$$

for any $q_0 \in \mathbb{R}$.

Proof. By Definition 7.1, $P\Phi_\gamma = \gamma\Phi_\gamma$. If (8.1) also held, then Φ_γ would be a nonzero simultaneous generalized eigenvector of Q and P, contradicting Theorem 2.8. □

Corollary 8.2 (Exact, not statistical, non-localization)**.** *For every nontrivial zero $\rho = \beta + i\gamma$, the associated exact momentum orbital is algebraically forbidden from being an exact position orbital. The obstruction is the identity $[Q, P] = iI$ and does not depend on an uncertainty variance or on a measurement model.*

Remark 8.3 (What is and is not quantized)*.* In the present rigorous model, γ is the generalized eigenvalue of the self-adjoint momentum P, while β labels the member $\mathcal{A}_\beta$ of the arithmetic operator family. The theory does *not* yet promote β to a second self-adjoint observable. This is deliberate: doing so without defining a domain and a compatible canonical structure would reintroduce a merely formal "hat on a variable".

9. Operator form of the Riemann Hypothesis

Theorem 9.1 (Exact reformulation of RH)**.** *The Riemann Hypothesis is equivalent to the following operator statement: for every $0 < \sigma < 1$ and every $\gamma \in \mathbb{R}$,*

$$\mathcal{A}_\sigma \Phi_\gamma = 0 \quad \Longrightarrow \quad \sigma = \frac{1}{2}. \tag{9.1}$$

Equivalently, all generalized momentum-null modes of the critical-strip family occur in the central member $\mathcal{A}_{1/2}$.

Proof. By Theorem 7.5, $\mathcal{A}_\sigma\Phi_\gamma = 0$ is equivalent to $\zeta(\sigma+i\gamma) = 0$ in the open critical strip. Therefore (9.1) is exactly the assertion that every nontrivial zero has real part $1/2$. □

Warning 9.2 (No proof of RH has occurred)*.* Theorem 9.1 is an equivalence, not a proof of (9.1). The exact Heisenberg theorem proves that a zero orbital cannot be simultaneously localized in Q and P; by itself it does not force the family parameter σ to equal $1/2$. Any argument claiming otherwise would require an additional theorem tying σ to the canonical pair.

10. Finite operators for exact numerical work

Definition 10.1 (Finite Möbius and alternating operators)**.** For $N \in \mathbb{N}$ and real σ, define bounded operators on all of $\mathcal{H}$ by

$$\mathcal{M}_{\sigma,N} = \sum_{n=1}^{N} \mu(n) n^{-\sigma} V_n, \tag{10.1}$$

$$\mathcal{A}_{\sigma,N} = \sum_{n=1}^{N} (-1)^{n-1} n^{-\sigma} V_n. \tag{10.2}$$

Proposition 10.2 (Finite Fourier multipliers)**.** *The operators in Definition 10.1 are bounded normal operators with Fourier multipliers*

$$m_{\sigma,N}(p) = \sum_{n=1}^{N} \mu(n) n^{-\sigma-ip}, \tag{10.3}$$

$$a_{\sigma,N}(p) = \sum_{n=1}^{N} (-1)^{n-1} n^{-\sigma-ip}. \tag{10.4}$$

Proof. Each is a finite linear combination of commuting unitary Fourier multipliers V_n. Equations (10.3) and (10.4) follow from (3.4). □

Remark 10.3 (Purpose of finite truncations). The finite operators contain no domain ambiguity and can be used to test spectral profiles numerically. Such experiments do not replace the closed operator definitions of Sections 4 and 5.

11. What has been established and what remains open

11.1. **Rigorous statements established in this notebook.** The following statements have been proved within the present framework.

(1) Q and P are self-adjoint on the exact domains (2.5) and (2.7), with the exact canonical commutator (2.9) on $\mathcal{S}(\mathbb{R})$.
(2) No nonzero tempered distribution can possess simultaneously exact Q and exact P; this is Theorem 2.8.
(3) The arithmetic translations V_n form a unitary representation of the multiplicative semigroup of positive integers.
(4) For $\sigma > 1$, the Möbius operator $\mathcal{M}_\sigma$ is bounded and invertible, and its inverse is the zeta operator $\mathcal{Z}_\sigma$ by the exact operator Möbius inversion identity (4.4).
(5) For every $\sigma > 0$, the alternating arithmetic operator $\mathcal{A}_\sigma$ has the closed maximal realization (5.11)–(5.12).
(6) In the open critical strip, nontrivial zeros are exactly generalized null momentum modes of $\mathcal{A}_\sigma$.
(7) Every such exact momentum mode is algebraically forbidden by Heisenberg from being an exact position mode.

11.2. **Open problem that would be needed for an RH mechanism.**

Question 11.1 (Central structural question)**.** Is there a property derivable from the Möbius-generated operator structure itself that distinguishes $\mathcal{A}_{1/2}$ from all $\mathcal{A}_\sigma$, $\sigma \neq 1/2$, and forces every generalized null momentum mode of the family to occur at $\sigma = 1/2$?

Remark 11.2. The exact Heisenberg obstruction is necessary for the intended non-localization interpretation, but it is not sufficient to answer Question 11.1. A successful next step must produce a new structural link between the family parameter σ and the canonical pair (Q, P); it cannot be obtained merely by rewriting $\zeta(s)$ in operator notation.

Question 11.3 (Possible strengthening)**.** Can the real coordinate $\sigma - 1/2$ be promoted to a genuinely defined self-adjoint observable on an enlarged Hilbert space, with a specified domain and an exact commutation law, while preserving the Möbius inversion construction of $\mathcal{Z}_\sigma$? Any such construction must be given before assigning a quantum interpretation to the real part of a zero.

Notebook History

Version 0.1: Exact Hilbert-space domains for Q and P, the unitary arithmetic translations V_n, bounded Möbius inversion for $\sigma > 1$, the closed critical-strip alternating operator for $\sigma > 0$, generalized zero orbitals, and the algebraic Heisenberg non-localization theorem assembled into a single framework.

References

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